

0.1 Section 6.1 - Intro to Counting

There are 2 highways from Palm Springs to Upland and 3 highways from Upland to Corona. How many ways can one travel from palm Springs to Corona if you must travel through Upland

	PU1	PU2
UC1		
UC2		
UC3		

$$2 \cdot 3 = 6 \text{ ways}$$

This is the fundamental theorem of Counting

Theorem 1 (Fundamental Theorem of Counting)

If there are a ways to do task A and b was of doing task B, then there are $a \cdot b$ ways to do task A and B

Example 1

Let's do some example of this

- How many 2 letter initials are there if...
 - The letters can be repeated?

$$\underbrace{26}_{\text{Letter 1}} \cdot \underbrace{26}_{\text{Letter 2}} = \boxed{676}$$

- The letters CANNOT be repeated?

$$\underbrace{26}_{\text{Letter 1}} \cdot \underbrace{25}_{\text{Letter 2}} = \boxed{650}$$

Since we can choose any letter for the first one, but the last one we can choose all but the one we just chose, so $26 - 1 = 25$

- How many 7 digit phone numbers do not start with 0 or 1?

$$\underline{8} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} = \boxed{8,000,000}$$

3. How many ways can 7 people line up for a picture?

(a) There are no restrictions

$$\underline{7} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 7! = \boxed{5040}$$

(b) Dina must be the first person in the photo

$$\underline{1} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 6! = \boxed{720}$$

(c) Dina is NOT the first person in the photo

$$\underline{6} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 6 \cdot 6! = \boxed{4320}$$

OR

ways Dina w/ first + # ways Dina NOT first = total

$$\begin{aligned} \# \text{ ways Dina NOT first} &= \text{total} - \# \text{ ways Dina w/ first} \\ &= 5040 - 720 \\ &= \boxed{4320} \end{aligned}$$

(d) Dina must stand next to Omar

We do this by treating Dina and Omar as one pair

But since there's, two ways of arranging Dina and Omar, Dina -> Omar and Omar -> Dina, we gotta multiply it every twice

$$2(\underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1}) = 2 \cdot 6! = \boxed{1440}$$

(e) Dina must not stand next to Omar

Dina next to Omar + # Dina NOT next to Omar = total

$$\begin{aligned} \# \text{ Dina NOT next to Omar} &= \text{total} - \# \text{ Dina next to Omar} \\ &= 5040 - 1440 \\ &= \boxed{3600} \end{aligned}$$

4. A bit string is a sequence of 0's and 1's. How many bit strings are there of length 7?

- (a) If there are no restrictions

$$\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^7 = \boxed{128}$$

- (b) Begin with 01

$$\underline{1} \cdot \underline{1} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^5 = \boxed{32}$$

We could also just say how many ways can we arrange a 5 length bit string since the first two are fixed (01)

- (c) Are palindromes, ie are the same when written forwards or backwards

$$\underbrace{\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2}}_{\text{anything}} \cdot \underbrace{\underline{1} \cdot \underline{1} \cdot \underline{1}}_{\text{fixed}} = 2^4 = \boxed{16}$$

5. How many ways can 5 people order a single scoops of icecream if the possible flavors are vanilla, coffee, chocolate, strawberry, or olive?

$$\underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} = 5^5 = \boxed{3125}$$

Since there's 5 flavors and 5 people

6. How many whole numbers less than 100 are...

- (a) Divisible by 3

$$\left\lfloor \frac{100}{3} \right\rfloor = 33 \# \text{ of possible ints div by 3}$$

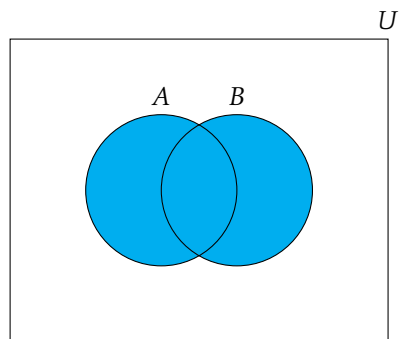
$$\left\lfloor \frac{100}{3} \right\rfloor + \underbrace{1}_{\text{including 0}} = \boxed{34}$$

- (b) div by 7?

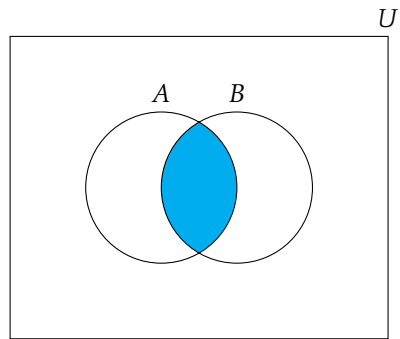
$$\left\lfloor \frac{100}{7} \right\rfloor + 1 = 14 + 1 = \boxed{15}$$

- (c) div by 3 OR 7?

This represents 3 OR 7, where A is all numbes div by 3 and B is same but for 7



This represents div by BOTH 3 and 7



Recall 1

$$A \cup B = A + B - A \cap B$$

Being divisible by both 3 and 7 just means being divisible by 21

$$A \cap B = \left\lfloor \frac{100}{3 \cdot 7} \right\rfloor + 1 = 4 + 1 = 5$$

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= 34 + 15 - 5 \\ &= \boxed{44} \end{aligned}$$

7. A die is rolled 5 times. How many ways can this happen if...

(a) No restrictions

$$\underline{6} \cdot \underline{6} \cdot \underline{6} \cdot \underline{6} = 6^5 = \boxed{7776}$$

(b) 4 is never rolled

$$\underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} = 5^5 = \boxed{3125}$$

(c) The number 4 is rolled at least once

ways 4 is rolled at least once + # never rolled = total

ways 4 is rolled at least once = total - # never rolled

$$= 7776 - 3125$$

$$= 6^5 - 5^5$$

$$= \boxed{4651}$$

8. A pair of dice is rolled. How many outcomes are possible if...

(a) No restrictions

$$\underline{6} \cdot \underline{6} = \boxed{36}$$

(b) The sum is 4

$$(1, 3), (3, 1), (2, 2)$$

so 3 ways

9. A coin is flipped 10 times. How many outcomes are possible if

(a) No restrictions

$$\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^{10} = \boxed{1024}$$

(b) At least 1 of the flips is heads

$$\begin{aligned}
& \# \text{ ways no heads} + \# \text{ ways at least 1} = \text{total} \\
& \# \text{ at least 1} = \text{total} - \underbrace{\# \text{ ways no heads}}_{\text{all tails}} \\
& = 2^{10} - 1 \\
& = \boxed{1023}
\end{aligned}$$

0.1.1 Multiples of n up to k

Theorem 2 (Number of Multiples of n up to k)

To find the number of positive integers divisible by n up to k is given by

$$\left\lfloor \frac{k}{n} \right\rfloor$$

This DOES NOT include 0, so for all whole numbers it would be

$$\left\lfloor \frac{k}{n} \right\rfloor + 1$$